

Memory Management - Summary

Learning Outcomes

- Understand memory hierarchy and OS roles in managing memory.
- Explain memory protection, address binding, and MMU.
- Differentiate logical vs physical addresses.
- Understand swapping, fragmentation, paging, and thrashing.

1. Memory Hierarchy & OS Responsibilities

- Registers > Cache > Main Memory (RAM) > Secondary Storage.
- Only RAM and registers are directly accessible by CPU.
- OS Tasks:
 - Track memory usage (free/allocated).
 - Allocate memory (contiguous/non-contiguous).
 - Translate addresses (logical -> physical).
 - Ensure protection (isolation between processes).

2. Memory Protection

- Prevents processes from accessing others' memory or OS.
- Base register: minimum legal address.
- Limit register: size of accessible memory range.
- If a process accesses outside its range -> error.

3. Address Binding

Three stages:

1. Compile Time - Absolute addresses known.

2. Load Time - Relocatable addresses converted when loaded.

3. Run Time - Addresses converted during execution (most flexible).

4. Logical vs Physical Address

- Logical Address: generated by CPU; a virtual view.
- Physical Address: actual location in memory.
- MMU (Memory Management Unit) translates logical -> physical.
- Dynamic relocation: Uses relocation register for real-time mapping.

5. Swapping

- Temporarily moves processes between RAM and disk.
- Enables multitasking with limited memory.
- Costly in terms of time (e.g., 4 seconds to swap 100MB in/out).

6. Contiguous Memory Allocation

- Memory divided into fixed/variable partitions.
- First Fit: first available block.
- Best Fit: smallest block that fits (minimizes waste).
- Worst Fit: largest block (leaves big free space).

Fragmentation

- Internal: Wasted space inside allocated block.
- External: Free memory exists but not in one block.
- Solutions: Compaction or non-contiguous allocation.

7. Paging

- Solves external fragmentation by breaking memory into:

- Pages (logical) and Frames (physical).
- Pages are mapped to frames via a Page Table.

Address Translation

- Address = Page number + Offset.
- TLB (Translation Lookaside Buffer) speeds up lookup.

Fragmentation in Paging

- No external fragmentation.
- Minor internal fragmentation in last page.

8. Virtual Memory & Page Replacement

- Virtual Memory: Processes can exceed physical RAM.
- Page Fault: Requested page not in RAM -> page is loaded.
- Page Replacement Algorithms:
 - FIFO, LRU, Optimal, Random, etc.

9. Thrashing

- System keeps swapping pages, doing little actual work.
- Caused by too many processes or too few frames.
- Prevention: Limit degree of multiprogramming.